

석사학위 논문

WBC YOLO-ViT:
YOLOv5와 Vision Transformer를 결합한 2
가지 방법 - 2단계 백혈구 검출 및 분류

WBC YOLO-ViT:
2 Way - 2 Stage White Blood Cell Detection
and Classification with a Combination of
YOLOv5 and Vision Transformer

2023년 8월

순천향대학교 대학원

미래융합기술학과

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이 논문을 석사학위 논문으로 제출함

2023년 6월

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Dedicated to my family.

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MS/FC 세르바스 아돌프 타리모. WBC YOLO-ViT: YOLOv5와 Vi-
20217183 sion Transformer를 결합한 2가지 방법 - 2단계 백혈구 검출 및
 분류.
 미래융합기술학과. 2023. 41p.
 지도교수: 우지영.

국 문 요 약

정확한 백혈구, 즉 백혈구의 감지와 분류는 다양한 질병의 진단 및 모니터링에서 중요한 역할을 합니다. 그러나 훈련된 전문가에 의한 수동 분류와 같은 기존 방법은 정확성, 효율성, 그리고 잠재적인 편향성 측면에서 수정되어야 합니다. 또한, 한정된 데이터, 해상도 노이즈, 불규칙한 형태, 그리고 다양한 소스로부터 다르게 변화하는 색깔 때문에 현미경 이미지를 이용하여 백혈구를 감지하고 분류하기 위해 딥 러닝 기법을 적용하는 것은 도전적입니다. 이 연구는 의료 영상에 구현된 백혈구 감지 및 분류에서의 도전 과제를 해결하기 위해 객체 감지와 비전 트랜스포머 모델을 통합하는 혁신적인 접근법을 제시합니다. 우리의 주요한 공헌은 기존 기법과 딥 러닝 방법의 한계를 효과적으로 극복하는 혁신적인 하이브리드 모델의 개발입니다. 제안된 모델은 16가지 유형의 백혈구 분류에서 0.99640의 뛰어난 정확도를 보여줍니다. 이 접근법은 조기 질병 감지와 치료 효과 모니터링을 포함한 임상 응용에 중요한 잠재력을 제공하며, 이로 인해 의료 전문가들이 더욱 정확하고

효율적으로 질병을 진단할 수 있게 합니다.

Keywords: 질병 감지, 질병 모니터링, 하이브리드 모델, 의료 영상, 물체 감지, 비전 변환기 모델, 백혈구 분류, 백혈구 감지, 딥 러닝.

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MS/FC Servas Adolph Tarimo. WBC YOLO-ViT: 2 Way - 2 Stage
20217183 White Blood Cell Detection and Classification with a Combination of YOLOv5 and Vision Transformer.
Department of Future Convergence Technology. 2023. 41p.
Advisor: Prof. Jiyoung Woo.

Abstract

Accurate detection and classification of white blood cells, otherwise known as leukocytes, play a critical role in diagnosing and monitoring various illnesses. However, conventional methods, such as manual classification by trained professionals, must be revised in terms of accuracy, efficiency, and potential bias. Moreover, applying deep learning techniques to detect and classify white blood cells using microscopic images is challenging owing to limited data, resolution noise, irregular shapes, and varying colors from different sources. This study presents a groundbreaking approach integrating object detection and vision transformer models to address the challenges in white blood cell detection and classification implemented in medical imaging. Our essential contribution is the development of an innovative hybrid model that effectively overcomes the limitations of traditional techniques and deep learning methods. The proposed model demonstrates an exceptional 0.99640 accuracy rate in classification of 16 types of white blood cells. This approach offers significant potential for clinical applications, including early disease detection and treatment efficacy monitoring, thereby enabling medical professionals to diagnose conditions more accurately and efficiently.

Keywords: Disease detection, Object Detection, Disease Monitoring, Medical Imaging, Vision transformer models, White Blood Cell Classification, Hybrid Model, White Blood Cell Detection, White Blood Cell Detection, Deep Learning

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Chapter 1

Introduction

1.1 White Blood Cell

Blood comprises various components, including sugars, hormones, proteins, minerals, carbon dioxide, and blood cells, which are categorized into three main types: red blood cells (RBCs), white blood cells (WBCs), and platelets. These cells, distinct in colour, shape, and structure, fulfil specific bodily roles. For example, while RBCs transport oxygen throughout the body, WBCs defend against infectious diseases and foreign bodies, and platelets ensure blood clotting, which is vital for stopping bleeding. The various types of WBCs in Fig. 1.1, namely monocytes, eosinophils, basophils, lymphocytes, and neutrophils [1], are instrumental in combating various infections. Nonetheless, their detection and classification remain challenging owing to the human body's RBC-to-WBC ratio of 600:1 [2] and complexity involved in accurately identifying the types of WBCs in a blood smear sample. Recently, there have been significant advances in deep learning techniques applied to medicine, underlining their potential in detecting and classifying WBCs in blood samples. For instance, to segregate WBCs from other blood cells, researchers have employed segmentation [3] methods such as clustering and computer vision techniques, including K-means clustering and edge detection algorithms [4]. Thresholding methods distinguish cells based on pixel intensity [5]. Furthermore, these methods are used for WBC detection. Notably, convolutional neural networks (CNNs) are remarkably advancing in direct feature extraction [6, 7, 8] from raw pixel data for cell classification.

Deep learning models operating in White Blood Cell (WBC) classification encounter two significant challenges. Both of these hurdles necessitate comprehensive and sophisticated solutions to unlock the full potential of deep learn-

ing in this domain and eventually enhance the accuracy of WBC classification. The first challenge arises from the limited availability of annotated datasets for realistic WBC images. This limitation becomes particularly pronounced when considering rare classes of WBCs. The scarcity of annotated datasets hampers the growth and evolution of deep learning techniques applied in the domain of hematology, as these models rely heavily on extensive datasets to learn and make accurate predictions.

Confronting this challenge demands the creation of larger and more diverse datasets that include a broad spectrum of WBC classes. These enriched datasets provide a more detailed and comprehensive foundation for deep learning models to train on, enhancing their learning potential and predictive accuracy. Deep learning models' second significant challenge is the existence of imbalanced class distributions within WBC datasets. This imbalance is characterized by certain classes having significantly fewer samples than others, skewing the representation of different WBC types.

Addressing this issue requires implementing advanced techniques to effectively manage class distributions. Sophisticated methods, such as resampling or adjusting class weights, can help ensure that all WBC classes are adequately represented in the learning process. This, in turn, improves the model's classification performance, particularly for rare WBC classes. By rising to meet these challenges by developing larger annotated datasets and applying advanced techniques to handle class imbalances, we can refine deep learning-based approaches to heighten the accuracy of WBC classification. Not only does this result in more precise disease diagnoses, but it also holds the potential to alleviate the workload on healthcare professionals by automating a significant portion of the blood analysis process. The road to overcoming these challenges may be steep, but the pursuit is crucial for advancing diagnostic healthcare and enhancing patient care.

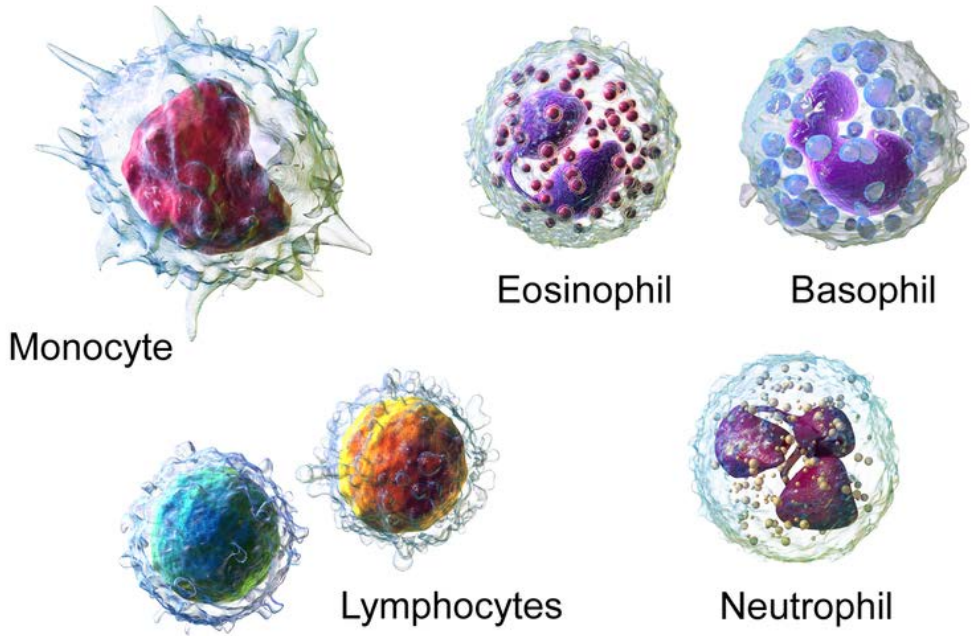


Figure 1.1: Various types of WBCs.

1.2 Motivation of the study

The motivation of the study was the ambition to advance the accuracy and efficiency of White Blood Cell (WBC) detection and classification through the innovative application of You Only Look Once (YOLO) and Vision Transformer (ViT) methods. These techniques present powerful potential for overcoming the existing challenges faced in the field, and our objective is to synergize their strengths in a uniquely designed, 2-way, 2-stage model. This model addresses the complex issues tied to object detection and classification in medical imaging, an area ripe for significant innovation.

This motivation stems from our deep-seated desire to spearhead a transformative shift in WBC detection and classification, positioning these as critical supports for our research. Further intensifying our motivation is the aspiration to contribute substantially to the medical field by investigating a broad spectrum of 16 distinct WBC types. This range includes rare classes, providing a more thorough examination than previous studies and enhancing our understanding

of these critical cells. To improve the medical field’s capacity to classify WBCs accurately and efficiently, our research plays a supreme role. Pursuing this research holds significant implications for the early diagnosis and treatment of various diseases. It is our hope that the fruits of our work will aid in improving patient outcomes and overall healthcare delivery..

1.3 Research Problem Definition

One of the primary challenges in the medical imaging domain resides in detecting and classifying white blood cells (WBCs). This difficulty is improved by the high Red Blood Cell (RBC) to WBC ratio in the human body, which stands at a staggering 600:1. The complexity involved in accurately identifying the types of WBCs within a blood smear sample further intensifies this challenge, making it a difficult task.

Our study focus in this difficult research problem. We aim to tackle the difficulties of accurately detecting and classifying a wide array of data categories within medical imaging, focusing our attention on WBCs. Over the years, machine learning and deep learning techniques have made inroads into medical applications, hinting at their significant potential in detecting and classifying WBCs in blood samples.

However, this potential is attended by a sequence of problems. The main issues we seek to address in this study are as follows:

1. **Limited Focus on Data Categories:** Many existing studies and techniques focus on a small number of classes with a large number of samples. This narrow focus overlooks the need for a comprehensive approach capable of detecting and classifying a wide range of WBC types, including rare classes that may be crucial for diagnosing certain diseases. The inability to accurately classify across a broad spectrum of WBC types is a major research problem this study seeks to address.
2. **Detecting and Classifying Irregular Shapes:** WBCs come in various

shapes and sizes, some of which can be quite irregular. Traditional detection and classification algorithms may struggle to handle these irregular shapes, leading to inconsistent and inaccurate results. Developing a robust method that can accurately identify and classify WBCs regardless of their shape is a significant challenge this research aims to tackle.

3. **Drawbacks of Manual Classification:** Manual classification of WBCs requires a considerable amount of time and expertise, making it impractical for large-scale use. Additionally, it is prone to human errors and inconsistencies due to subjective interpretation. Therefore, developing an automated system that can outperform manual methods in terms of speed, accuracy, and consistency is an important research problem.
4. **Imbalanced Dataset Issue:** A related problem is the presence of imbalanced datasets in medical imaging, where certain classes have significantly fewer samples than others. This imbalance can lead deep learning models to be biased towards classes with more instances, resulting in a decline in classification performance on rare classes.
5. **Lack of Annotated Datasets:** There is a scarcity of well-annotated datasets, particularly those including rare classes of WBCs. This shortage limits the training and effectiveness of deep learning models, hindering advancements in WBC detection and classification.

To solve these problems, the focus of this research is to use YOLO and vision Transformer for detecting and classification of White blood cells. We propose a 2-way 2-stage model that combines the You Only Look Once (YOLO) and Vision Transformer (ViT) methods to enhance the detection and classification of WBCs, including those of irregular shapes and rare classes, while outperforming traditional manual methods.

1.4 Research Contributions

The main contributions of this study are as follows:

1. **Two-way object detection:** We propose a 2-way approach to use two types of images, including WBC and nucleus images, while previous research used only blood cell images or WBC-focused images.
2. **Hybrid architecture of object detection and classification:** We present a hybrid architecture that combines the strengths of YOLO (fast object detection) and ViT (powerful image representation capabilities), effectively integrating convolutional and transformer-based approaches for object detection and classification tasks in medical imaging.
3. **Comprehensive multi-class dataset:** Our study stands out from prior research in terms of the wide range of 16 distinct WBC investigated, including rare classes.

1.5 The Overall Organization of the Thesis

The current thesis has comprised of seven chapters and it is organized as follows.

Chapter 1 This chapter introduces White Blood Cells, discusses the challenges encountered in their functioning, highlights the motivation that drives the study, defines the research problem, and outlines the research contributions.

Chapter 2 This chapter provides a concise overview of the research context and motivation. Firstly, we will explore the background by discussing the techniques used to detect and classify various types of White blood cells. Through this exploration, the chapter aims to establish a hybrid method for detection and classification.

Chapter 3 This chapter provides a comprehensive review of existing literature on YOLO, CNN-Based WBC Classification, and Vision Transformer. It elucidates various architectures and techniques employed for the detection and classification of White Blood cells, along with the extracted features.

Chapter 4 In the methodology chapter, we outline our approach using a 2-way 2-stage framework, which consists of object detection and classification. To begin, we introduce the detection and classification of white blood cells (WBC) using YOLO. We commence by elaborating on the preprocessing steps, followed by a discussion on classification utilizing vision transformer.

Chapter 5 In this chapter, we conducted an experimental study and obtained the results. The obtained results are then compared to the findings in the previous literature, highlighting the differences and similarities. Additionally, the chapter delves into the ablation analysis conducted to further analyze and interpret the results.

Chapter 6 In this chapter, we will present a comprehensive discussion and outline the remaining challenges.

Chapter 7 In conclusion, this chapter comprehensively presents the entirety of the work while also providing insights into future directions.

Chapter 2

Background

2.1 Background

Accurate and efficient detection and classification of white blood cells (WBCs) are essential in analyzing and monitoring various illnesses. Historically, the process has been carried out manually by trained medical professionals who inspect and interpret microscopic images of blood samples. These conventional methods, though established, present several limitations, such as potential biases, inconsistencies due to human error, and significant time expenditure. With technological advances and the quickness of current healthcare needs, it has become increasingly clear that these standard methods must be inspected and repaired.

To handle these issues, there has been a modification towards leveraging artificial intelligence, and more intense learning techniques, to automate and improve the WBC detection and classification process. Deep learning has shown great promise in various domains, including medical imaging, owing to its ability to learn intricate patterns from large amounts of data. However, applying deep learning techniques in the context of WBC detection from microscopic images is challenging.

Recently, convolutional neural networks have been widely adopted for image-based tasks due to their capability to handle pixel-level information. CNNs excel at learning hierarchical features from images, making them practical for object detection and recognition tasks. However, they are somewhat limited when it comes to modelling the spatial relationships between different parts of an image, which is a crucial aspect of WBC classification due to the morphological diversity of these cells.

On the other hand, Vision Transformer (ViT), a newer addition to the deep

learning repertoire, addresses some of the limitations of CNNs. Unlike CNNs, ViT does not explicitly focus on local features. Instead, it uses self-attention mechanisms to model changes in local images and their long-range dependencies, effectively capturing global contextual information within images. This makes ViT particularly suited to handle the variability and complexity associated with WBC images.

However, the deployment of these techniques in medical imaging is plagued by a range of problems, including the limited availability of annotated data, resolution noise, irregular shapes of the WBCs, and varying colors resulting from different staining and lighting conditions. These challenges necessitate a fresh approach that integrates the strengths of both CNN and ViT models.

A groundbreaking approach that seeks to navigate these issues integrates object detection techniques, such as You Only Look Once (YOLO) and Vision Transformer models. This novel two-stage approach starts with object detection to identify the regions containing WBCs, effectively addressing the challenge of low occurrence of WBCs in blood samples. This stage ensures precise localization of the WBCs, which is followed by a classification stage. Here, features are extracted from the identified regions and fed into a pre-trained Vision Transformer model for accurate classification.

This innovative hybrid model has shown exceptional performance in classifying 16 types of WBCs, including rare classes, with an impressive accuracy rate. Such advancements are a boon for clinical applications, as they enable more accurate and efficient diagnoses, promising to significantly contribute to early disease detection and monitoring of treatment efficacy. The approach stands as a testament to the potential of deep learning in advancing our understanding of the immune system and enhancing the tools available for disease diagnosis.

Chapter 3

Related Works

3.1 Related Work

We review three areas of study: YOLO, CNN-based WBC classification, and ViT. The YOLO model was examined to understand its application to extraction of WBC and nucleus images from blood cell images. CNN (Convolutional Neural Network)-based WBC models were reviewed to improve the performance of existing models in scenarios where numerous classes, including rare classes, are present. ViT model, which utilizes self-attention mechanisms, was explored as a promising approach for WBC classification. Table 3.1 provides an overview of relevant WBC object detection and classification studies.

3.1.1 YOLOv5 Application

Recent advancements and progress in machine learning and deep learning method have caused a research surge focusing on detection of blood cells. Rohaziat et al. [9] applied the cutting-edge YOLOv5 detection model to identify various WBCs, using a dataset of 2,800 images obtained from Kaggle.com, and achieved a mean average precision (mAP) of 99.42% with the YOLOv5m model. Furthermore, Shinde et al. [10] improved YOLOv5's accuracy over its predecessors by employing automatic detection, segmentation, and counting of blood cells using smear images, aiming to facilitate real-time diagnosis. Their method focused on detecting three significant blood cell categories, achieving 93% mAP. Similarly, to simultaneously capitalize on segmentation and classification, several studies introduced frameworks for WBC classification and segmentation [11, 12, 13, 14]. Yao et al. [15] used Faster RCNN and Yolov4 on an enhanced Blood Cell Count Dataset (BCCD) to classify WBCs and proposed an effi-

cient detection method that combines segmentation [16] and target recognition, achieving accuracy rates of 96.25% and 95.75%. Zhao et al. [17] proposed a YOLOv5-based blood cell detection and counting method, training the model on BCCD and achieving a high mAP of 0.96 based on 46 samples. In another approach, Sarkar et al. [18] improved the YOLOv5 model to accurately detect small objects such as WBCs, a critical component in medical diagnosis owing to their vital immune system role. They reached an accuracy of 96.50%. Luo et al. [19] combined YOLOv5 with ResNet to detect chest abnormalities in X-ray images, achieving 25.4% higher mAP than the faster RCNN detection method.

3.1.2 CNN-Based WBC Classification

With the advent of deep learning techniques, CNN-based methods have been extensively utilized for classification of WBCs. They have proven to be highly effective in extracting relevant features from WBC images, achieving remarkable classification performance. Khan et al. [22] introduced MLANet-FS. This model integrates AlexNet [26] with a feature selection strategy tailored for classification of different WBC types. Girdhar et al. [24] advocated for the significance of WBC analysis and introduced a CNN approach for WBC classification, achieving a high accuracy of 98.55% on a Kaggle dataset. Heni et al. [25] proposed the use of the EK-means algorithm and data augmentation techniques to classify WBC subtypes, yielding a 96.24% accuracy. Wijesinghe et al. [27] emphasized the importance of WBC measurement and proposed a computer vision-based method based on k-means clustering for WBC detection and localization and VGG-16 [28] classifier for classification, achieving an accuracy of 95.89%. Akalini et al. [29] proposed real-time WBC detection using YOLOv5s, YOLOv5x, and Detectron2 R50-FPN pre-trained models, optimizing the activation function and architecture arrangement and achieving a maximum accuracy rate of 98%. Chien et al. [30] used Faster R-CNN and CNN to identify and categorize immature WBCs, achieving approximately 90% accuracy. Jung et al. [31] proposed a CNN-

Table 3.1: Summary of the state-of-the-art models for WBC detection and classification

Author (Year)	Number of Class	Number of Samples	Model tion(Detection fication)	Descrip- / Classi-	Object detection perfor- mance: mAP	Classificat perfor- mance: Accuracy
Shinde et al. [10] (2021)	-	864	YOLOv5		0.9300	-
Yao et al. [15] (2021)	4	12,500	Faster RCNN	and YOLOv4	-	0.9625- 0.9575
Rohaziat et a. [9] (2022)	4	2,800	YOLOv5		0.9942	-
Zhao et al. [17] (2022)	4	12,500	YOLOv5		0.9600	-
Razzak et al. [20] (2017)	5	64,000	CNN combined with extreme learning machine (ELM)		-	0.9868
Özyurt et al. [21] (2020)	4	12,494	Ensemble of CNN models with MRMR and ELM classifier		-	0.9603
Khan et al. [22] (2021)	4	12,444	CNN model with feature selection strategy (MLANet-FS)		-	0.9912
Baghel et al. [23] (2022)	4	12,500	CNN model		-	0.9851
Girdhar et al. [24] (2022)	4	12,444	CNN model		-	0.9855
Heni et al. [25] (2023)	4	12,444	CNN model with EK-means algorithm		-	0.9624

based WBC classification method namely, W-Net, achieving 97% accuracy on a large-scale dataset. Özyurt et al. [21] introduced a fused CNN model employing pre-trained architectures for WBC detection. Razzak et al. [20] used a combination of deep contour-aware CNN and extreme machine learning for WBC segmentation and classification.

Overall, computer vision and deep learning techniques have demonstrated significant potential for WBC classification, a crucial step for diagnosing pathological conditions. Previous research highlights the importance of data augmentation, segmentation, and localization techniques in improving the accuracy of WBC classification models, assisting healthcare professionals in diagnosing and treating illnesses.

3.1.3 Vision Transformer (ViT)

The original transformer architecture has been successful in natural language processing applications, prompting efforts to adapt it for visual tasks, such as image classification [32], semantic segmentation [33], and object detection [34]. The ViT developed by Dosovitskiy et al. [32] demonstrates the effectiveness of a transformer-based approach in image classification by dividing images into tokens and using transformer layers to capture global relationships. Mechanisms similar to transformers' self-attention [35] are employed in other models, such as squeeze and excitation [36], selective kernel [37], and split-attention networks [38]. Other studies explored implementing a localized self-attention mechanism using a shifted window strategy or adopting a pyramid structure for diverse feature maps. Liu et al. [39] introduced a shifted window strategy to implement self-attention within a localized window, alternating between modeling in and cross-window connections. Heo et al. [40] adopted a pyramid structure resembling CNNs to generate diverse feature maps through spatial dimension reduction based on the pooling configuration inherent to a convolutional layer. Yuan et al. [41] enhanced the tokenization in ViT by carefully creating tokens through

a soft split approach, recursively combining adjacent tokens into a single pass to improve local structure modelling. Sengar et al. [42] proposed an automated deep learning framework for malaria diagnosis using microscopic blood cell images, employing a transformer-based neural network. Tanzi et al. [43] compared the ViT architecture with a classic CNN model for bone fracture classification. Bazi et al. [44] implemented ViT for classification of remote-sensing scenes. Manzari et al. [45] developed the MedViT model, a hybrid CNN-transformer architecture for medical image analysis, addressing computational complexity and adversarial attacks. Ren et al. [45] proposed a combined model comprising ResNet-50 and a transformer for classifying COVID-19 lesions in lung CT/X-ray scan images. Uparkar et al. [46] compared ViT with a hybrid CNN approach for lung disease detection. Overall, transformer-based approaches have proven to be versatile in the medical domain.

In this review, we discovered three research gaps. First, object detection methods (e.g., YOLO and Faster R-CNN) accomplish high accuracy in detecting WBCs from images. However, in terms of classification performance, they may achieve a different level of accuracy than that of CNN-based classification methods. Object detection methods excel in localizing and identifying WBCs, while classification methods offer more detailed analysis and categorization. Second, CNN-based classification models have achieved superior performance in classifying major classes of WBCs. They leverage the power of CNNs to automatically learn discriminative features from WBC images, thereby accomplishing high accuracy in classification of different types of WBCs. However, for real-world deployment, it is crucial to develop models that perform well on rare classes of WBCs as well. Third, ViT, which has gained prominence in image-based deep learning applications, has yet to be adopted for WBC classification. ViT models utilize self-attention mechanisms to capture global and local relationships contained in images; therefore they are promising for WBC classification, where images are similar to each other and may contain rare classes.

Chapter 4

Proposed Method

4.1 Proposed Method

We present a 2-way 2-stage framework comprising of object detection and classification. Object detection consists of 2-way nucleus detection and WBC detection. Our core objective is to conduct a comparative analysis between WBCs and their respective nucleus. Following the detection stage, we implement the ViT for the classification process. The proposed model is illustrated in detail in the flowchart presented in Fig.4.1.

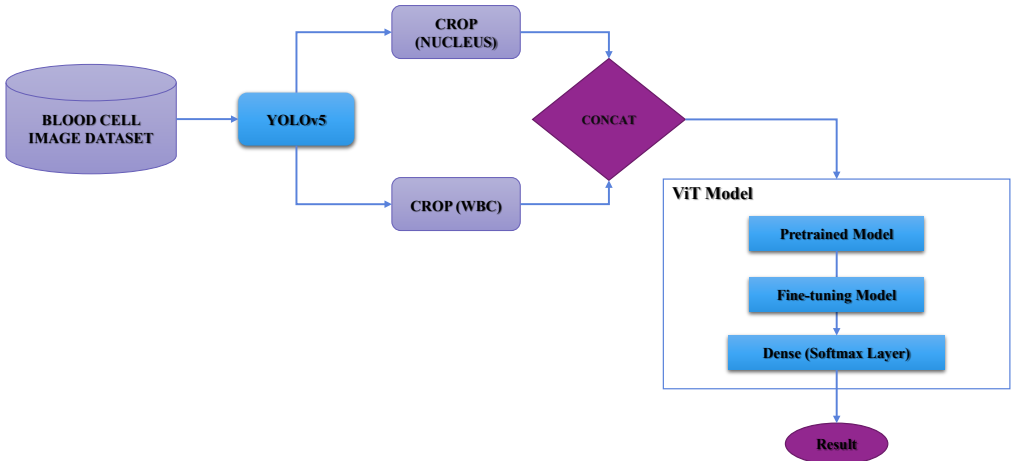


Figure 4.1: Proposed model, YOLO - ViT in a combination framework for detection and classification of WBCs.

4.1.1 Detection and Classification Model on WBC (YOLOv5)

YOLOv5 comprises three main components: the backbone, neck, and head, each serving a distinct purpose. The backbone extracts features from input images to process them hierarchically and extract multi-level features for detection and classification tasks [47]. The neck, an intermediate layer connecting the backbone and detection head, aggregates and refines features creating a multi-scale feature map that combines low and high-level features [48]. The head predicts the location of bounding boxes and classifies objects within them. It has multiple convolutional layers and three output layers for different scales, predicting class probabilities, determining box coordinates, and objectness scores. Traditionally, manual microscopic examination of blood smears visually assesses the shape and characteristics of WBCs and their nucleus. These features include WBC characteristics, such as cell size, shape, and texture, and nucleus characteristics, such as size, shape, and chromatin distribution [49]. Our model automates the key observation, traditionally conducted manually, through the 2-way classification to focus on and compare WBCs and their nucleus. The nucleus contains crucial information, such as shape, size, and chromatin patterns, that are vital in accurately characterizing and classifying WBCs. By cropping the nucleus, researchers can isolate and focus on these distinctive nuclear features, allowing for more precise analysis and classification of different WBC subtypes [50]. A bounding-box tool was used to create precise labels during data preparation. Then, the labelled training images were processed through the backbone, neck, and head components to generate the desired output. Subsequently, the weights obtained from training were utilized for testing and inference purposes. To capture both the WBC and nucleus regions accurately, the crop function was employed to capture them. Finally, the confidence function filtered out images with low scores, allowing the model to focus on the areas of interest contained in the image. This process is depicted in Fig. 4.2.

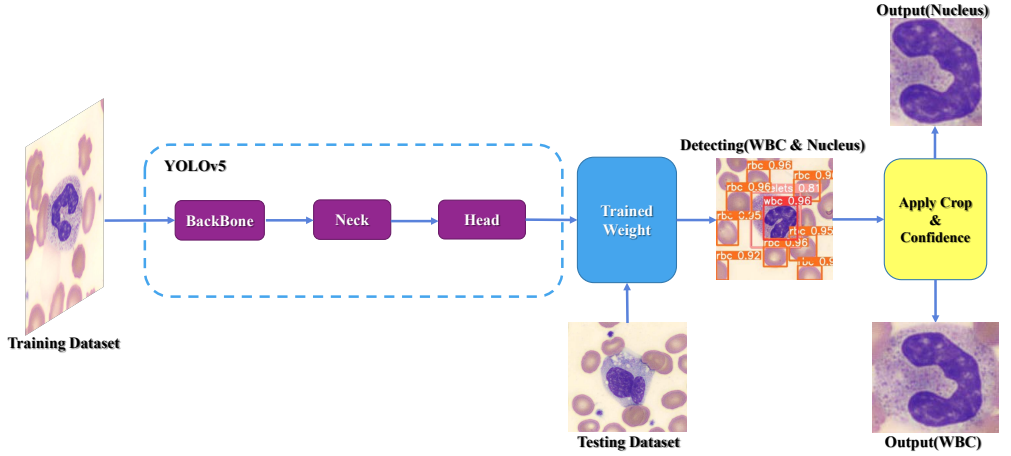


Figure 4.2: Nucleus and WBC detection using the YOLOv5 model.

4.1.2 Vision Transformer for Classification Model on WBC and nucleus

We leveraged ViT [32] as a popular pre-training model. ViT uses a unique process of image data handling, comprising patch-based image processing, sequence representation, and a self-attention mechanism. Unlike a CNN, which processes small, overlapping portions of images, ViT divides images into non-overlapping, equal-sized patches. It views the entire image holistically, disregarding hierarchy, which allows global context consideration. After dividing images into patches, ViT turns these 2D patches into a sequence of 1D vectors, similar to word sequencing implemented in natural language processing (NLP). ViT retains positional information through additional encoding, so that the model can understand the context broadly and capture long-range dependencies. ViT also employs a self-attention mechanism [35], assigning weights to each patch based on its importance with respect to others. This mechanism helps ViT grasp complex dependencies across the entire image, providing a superior understanding of the image context compared to that achieved with CNNs. ViT’s innovative approach enhances the understanding of detailed image data; consequently, it often outperforms CNNs in tasks such as WBC and nucleus detection. The ViT

model has two main variants: B (Base) and L (Large), which denote the size and complexity of the model. The number following the variant (16 or 32) denotes the patch size used to divide the input provided to the transformer apriori. For instance, a patch size 16 means that each image is divided into 16x16 pixel patches, while a patch size 32 would divide the image into 32x32 pixel patches. The original research conducted by Dosovitskiy et al. [32] titled "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale" first introduced this architecture. They used 16x16 patches and achieved state-of-the-art performance on several image classification benchmarks. The patch size considerably influences the performance, affecting the level of detail the model captures and its computational efficiency. When smaller patches are chosen, the model can extract detailed information from images, which is advantageous for tasks that require identifying subtle patterns. However, this increases the computational load resulting from the complex nature of the self-attention mechanism within the transformer model. By contrast, larger patches dismiss fine details but recognize broader patterns within the data. This choice reduces the number of tokens, decreasing the computational cost. The choice between smaller and larger patch sizes depends on the task specific requirements and the data. If obtaining intricate details is crucial for image classification, smaller patches might be a better choice despite their higher computational demands. Conversely, larger patches could provide sufficient information while enhancing computational efficiency if the task focuses more on general features and patterns. Self-attention demands high computational resources owing to its quadratic complexity associated with the sequence length, especially for larger images or smaller patches. Fig.4.3 shows a visual representation of the basic ViT architecture. To highlight the effectiveness of ViT in the context of WBC classification, we conducted a comparative analysis using CNNs as the comparison model. This study used a pre-trained CNN model i.e., ResNet50 [51]. Fig.4.4 depicts the basic CNN architecture.

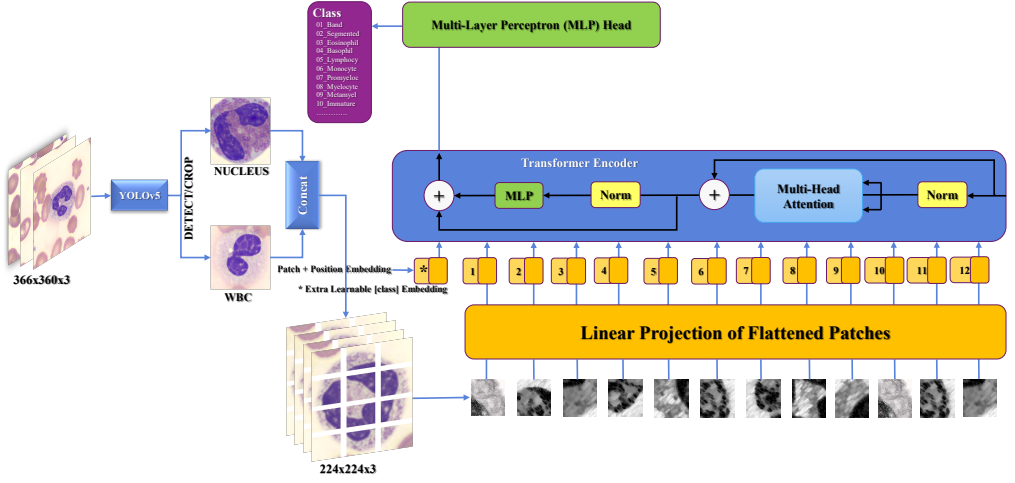


Figure 4.3: Proposed Our 2-way 2-stage WBC classification model using YOLO and ViT.

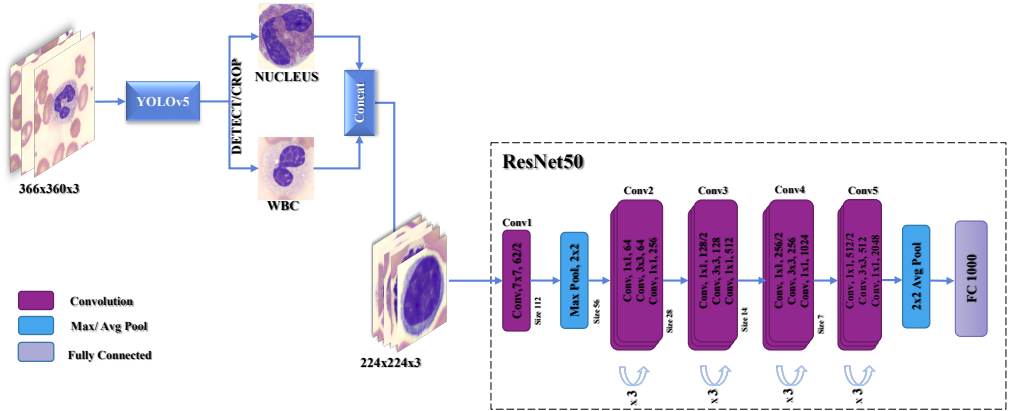


Figure 4.4: Comparison model using ResNet50.

Chapter 5

Experiments and Results

5.1 Experiments

5.1.1 WBCs Dataset

We propose the use of a dataset consisting of 4,445 labelled blood cell images. This dataset was collected from January 2022 to May 2022 in Soonchunhyang University Bucheon Hospital in South Korea. This study was approved by the Institutional Review Board of the Soonchunhyang University Bucheon Hospital (SCHBC 2022-12-015). Before conducting the intensity unification phase, the images had an original size of 360 x 366 on 3 RGB channels. They were obtained from blood films taken at Soonchunhyang University Hospital in South Korea. We aimed to classify each image into one of 16 different cell types, including Artifact, Large granular lymphocyte, Promyelocyte, Immature cell, Eosinophil, Lymphocyte, nRBC, Monocyte, Lymphocyte variant form, Segmented neutrophil, Smudge cell, Basophil, Band neutrophil, and Giant platelet. Table 5.1 lists the number of images per class. A sample from the dataset used in this study is displayed in Fig. 5.1.

Table 5.1: Confidence Scores per Class After Cropping

Class Name	Sample	Nucleus Conf	WBC Conf
Band neutrophil	328	0.90	0.80
Segmented neu- trophil	533	0.90	0.80
Eosinophil	510	0.80	0.90
Basophil	219	0.80	0.80
Lymphocyte	260	0.80	0.90
Monocyte	322	0.70	0.70
Promyelocyte	82	0.30	0.50
Myelocyte	93	0.60	0.80
Metamyelocyte	131	0.50	0.50
Immature cell	257	0.80	0.90
Lymphocyte vari- ant form	267	0.80	0.90
Large granular lymphocyte	129	0.80	0.80
Smudge cell	262	0.80	0.80
Artifact	290	0.70	0.80
nRBC	251	0.90	0.90
Giant platelet	511	0.80	0.90
Total Images	4,445	0.74	0.76

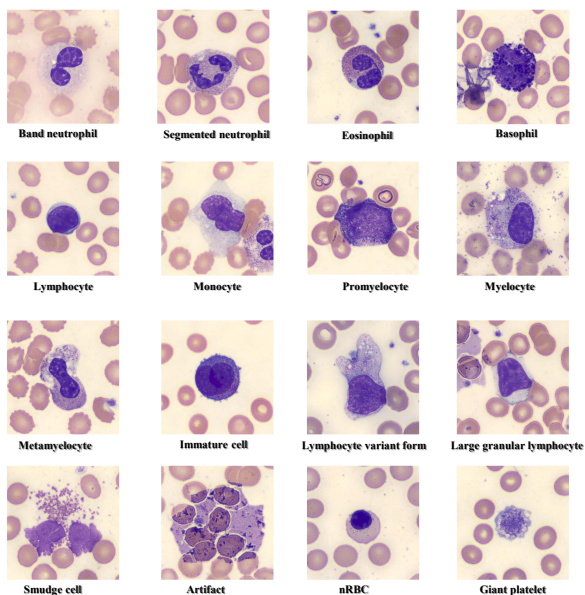


Figure 5.1: WBC samples taken from 16 classes.

5.1.2 WBC and Nucleus Detection

YOLO generates the bounding-box coordinates and corresponding confidence scores. The confidence score is calculated by multiplying the probability that the box contains an object and intersection union between the predicted box and real object. In an experiment specifically focused on detecting and classifying WBCs and their nuclei, our model achieved an average confidence score of 0.30 and maximum of 0.90 in different classes. Table 5.1 shows WBC and nucleus class names, number of samples per class, and confidence score. These results indicate that our model provides a high degree of assurance in accurately identifying and categorizing WBCs and their nucleus under certain circumstances, reaching up to a 30% and 90% confidence scores, respectively. Promyelocyte exhibits the lowest confidence in nucleus detection because the nucleus and WBC are not separated clearly. Metamyelocyte exhibits the second lowest confidence because its WBC and nucleus shapes are slanted peanut oval. This peanut oval shape causes a low IoU between the rectangle-shape bounding box and real image. Object detection models can also be used for classification tasks when the images are annotated with objects from different classes. Table 5.2 lists the evaluation metrics resulting from the object detection model for the classification task conducted on each category of WBCs and its respective nucleus. This table highlights the effectiveness and accuracy in identifying and categorizing various classes of WBCs, along with the detection of their nucleus, using the implemented object detection methods. Overall, object detection attains a lower accuracy than the classification models found in the literature review. In addition, the classification based only on the nucleus attains a lower performance than that considering WBCs. Figs. 5.2 and 5.3 display WBC and nucleus cropped images of the samples. Given that the object detection model struggles with the classification task, we combined the cropped WBC and nucleus images to proceed with the classification task. Subsequently, we divided the paired WBC and nucleus dataset into 3,600 images for training, 400 images

for validation, and 445 images for testing. These portions were subsequently fed into the ViT model for classification purposes.

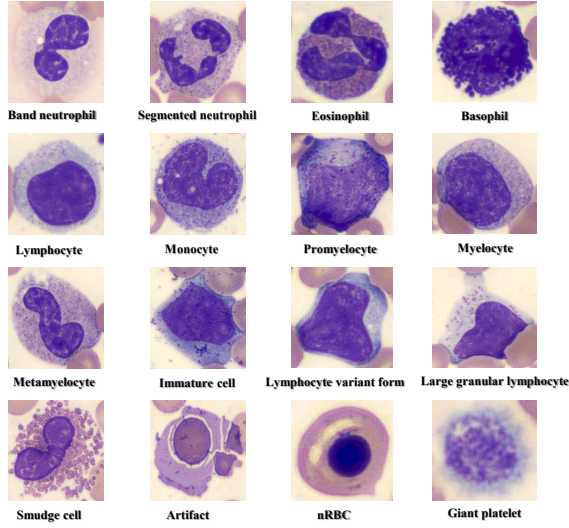


Figure 5.2: WBC cropped samples taken from 16 classes.

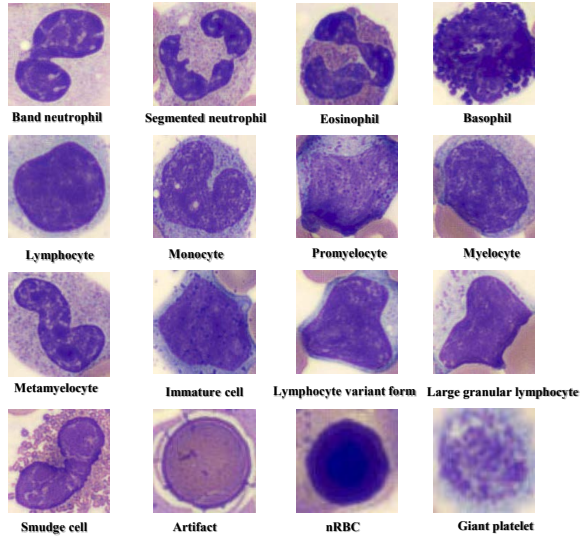


Figure 5.3: Nucleus cropped samples taken from 16 classes.

Table 5.2: Results of object detection

Class	Name	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	All
WBC	Precision	0.886	0.969	0.795	0.940	0.859	0.947	0.744	0.840	0.859	0.846	0.823	0.900	0.592	0.556	0.701	0.800	0.816
	Recall	1.00	1.00	0.97	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.989
	mAP@0.5	0.986	0.976	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.864	0.995	0.995	0.995	0.985
Nucleus	Precision	0.712	0.913	0.926	0.870	0.698	0.926	0.523	0.700	0.843	0.826	0.813	0.841	0.606	0.325	0.649	0.582	0.734
	Recall	1.00	1.00	0.97	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.892	1.00	1.00	0.936
	mAP@0.5	0.995	0.966	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.882	0.535	0.995	0.995	0.957

5.1.3 Classification

The 10-fold cross-validation technique was effortlessly applied in training, validation, and testing to assess unbiased performance. In addition, this division helped minimize the risk of overfitting the model to the data.

1. **ViT model setting:** The ViT model was used to conduct experiments after pre-training it on the ImageNet21k dataset at various sizes. The model architecture includes a sequence of self-attention layers, followed by a feed-forward network with 16 and 32 layers, hidden layers ranging from 768 to 1024. During pre-training, the input images were resized to 224x224 using 3 channels. For a specific image classification task, the pre-trained ViT model was fine-tuned on a smaller dataset namely, ImageNet2012, using cross-entropy loss as the fine-tuning objective and Adam optimizer with a fixed learning rate of 0.0001. The model was trained for a total of 20 epochs. The ViT model was trained on 8 different data-folds, validated on 1 data-fold, and tested on the remaining fold. We selected a random set of testing data from a set-aside testing sample to evaluate the model performance on each fold. This process was repeated ten times to evaluate the ViT model’s robustness and generalizability on the given image classification task.
2. **CNN model setting:** We implemented pre-trained CNN models by processing our dataset through various layers, thereby optimizing their established capabilities. A weight decay (L2) of 0.01 was used the regularization technique, imposing a penalty on the loss function for large weights and influencing the layer’s kernel and softmax activation function. The model underwent fine-tuning over 20 epochs with a 0.001 learning rate. To ensure compatibility, the images were resized to 224 x 224, encompassing three channels.

5.2 Results

5.2.1 Ablation Analysis

Table 5.3 lists the results obtained from 10-fold cross-validation. We compared the performance of two pre-trained models, namely, ViT and ResNet50. We employed various metrics, such as accuracy, precision, recall, and F1 score, to evaluate the classification model. Accordingly, we observed that the ViT outperformed the CNN model. ViT exhibited varying performance levels according to the number of patches and model size. The large model outperformed the based model, and implementing less patches were better than, implementing more patches. Fig. 5.4 displays the confusion matrices of the proposed model. The best model attains an accuracy of 99.640% after only 20 epochs for 16 classes, including rare classes. The results demonstrated improved accuracy and specificity, particularly for challenging WBC types such as Artifact, Large granular lymphocyte, Promyelocyte, Immature cell, Eosinophil, Lymphocyte, nRBC, Monocyte, Lymphocyte variant form, Segmented neutrophil, Smudge cell, Basophil, Band neutrophil, and Giant platelet. However, the identification of Myelocyte and Metamyelocyte remained difficult. We conducted an ablation analysis to evaluate the strengths of the proposed model compared to existing models. First, we designed a model without object detection, where ViT and ResNet50 were trained solely on blood cell images. We built a model in the second ablation analysis using only WBC images, excluding extra nucleus images. Table 5.4 lists the results without object detection. Note that the performance of both the ViT and ResNet50 was poor without the integration of object detection. Interestingly, ViT’s performance decreased by 4.1%, but ResNet50’s performance decreased as much as 6.3%. Furthermore, Table 5.5 elaborates on the performance of ViT, indicating that its effectiveness decreased by an additional 0.057 compared to previous results when we applied a cropping mechanism to isolate WBCs, removing the nucleus. In this case, ViT with a base configuration and 16 patches still performed well with respect to other results. This might

be caused by data simplicity rather than WBC and nucleus concatenation. ViT and ResNet50 models showed limitations when we used only the cropped WBC images. We observed that the performance decrease was more significant when object detection was excluded than when nucleus images were. However, note that the ViT model exhibited only a slight performance decrease in both ablation analyses. According to these ablation analyses, we further claim robustness and effectiveness of the proposed model. Despite the absence of object detection and exclusion of nucleus images, the ViT model demonstrated robustness and maintained relatively high performance. However, our aim was to highlight the importance of object detection in our model, as its exclusion led to a substantial performance decrease. Overall, our ablation analysis reinforces the advantages and strengths of the proposed model, underlining the significance of object detection and robustness of the ViT architecture in WBC analysis. Our model has the following strengths. In the first stage of the 2-stage model, object detection is performed to localize and identify the regions of interest (ROIs) containing the WBCs. This step helps in accurately localizing the WBCs contained in the image, reducing the risk of focusing on other cells. In the second stage, the ROIs detected in the previous step are further analyzed by the classification model. Using a dedicated classification model, the features extracted from the ROIs can be more effectively utilized for accurate classification.

Table 5.3: Performance of the proposed, 2-way 2-stage model with 10-fold cross-validation

Model	Accuracy	Precision	Recall	F1 Score
ViT_L16	0.99640	0.99375	0.99332	0.99271
ViT_L32	0.99595	0.99499	0.99441	0.99433
ViT_B16	0.99506	0.99254	0.99170	0.99167
ViT_B32	0.99326	0.98917	0.98718	0.98797
ResNet50	0.99258	0.98946	0.98737	0.98806

Table 5.4: Performance of the 2-stage model with 10-fold cross-validation without object detection

Model	Accuracy	Precision	Recal	F1 Score
ViT_L16	0.95505	0.94407	0.94281	0.94042
ViT_L32	0.93885	0.91703	0.91683	0.91374
ViT_B16	0.94291	0.93644	0.94022	0.93594
ViT_B32	0.93528	0.92637	0.92601	0.92422
ResNet50	0.92921	0.91901	0.92759	0.91897

Table 5.5: Performance of the 1-way 2-stage model with 10-fold cross-validation without extra nucleus images

Model	Accuracy	Precision	Recal	F1 Score
ViT_L16	0.93842	0.92187	0.91312	0.91304
ViT_L32	0.94313	0.93089	0.92922	0.92649
ViT_B16	0.94426	0.94041	0.94382	0.93985
ViT_B32	0.94089	0.93444	0.93548	0.93244
ResNet50	0.93460	0.93253	0.92786	0.92722

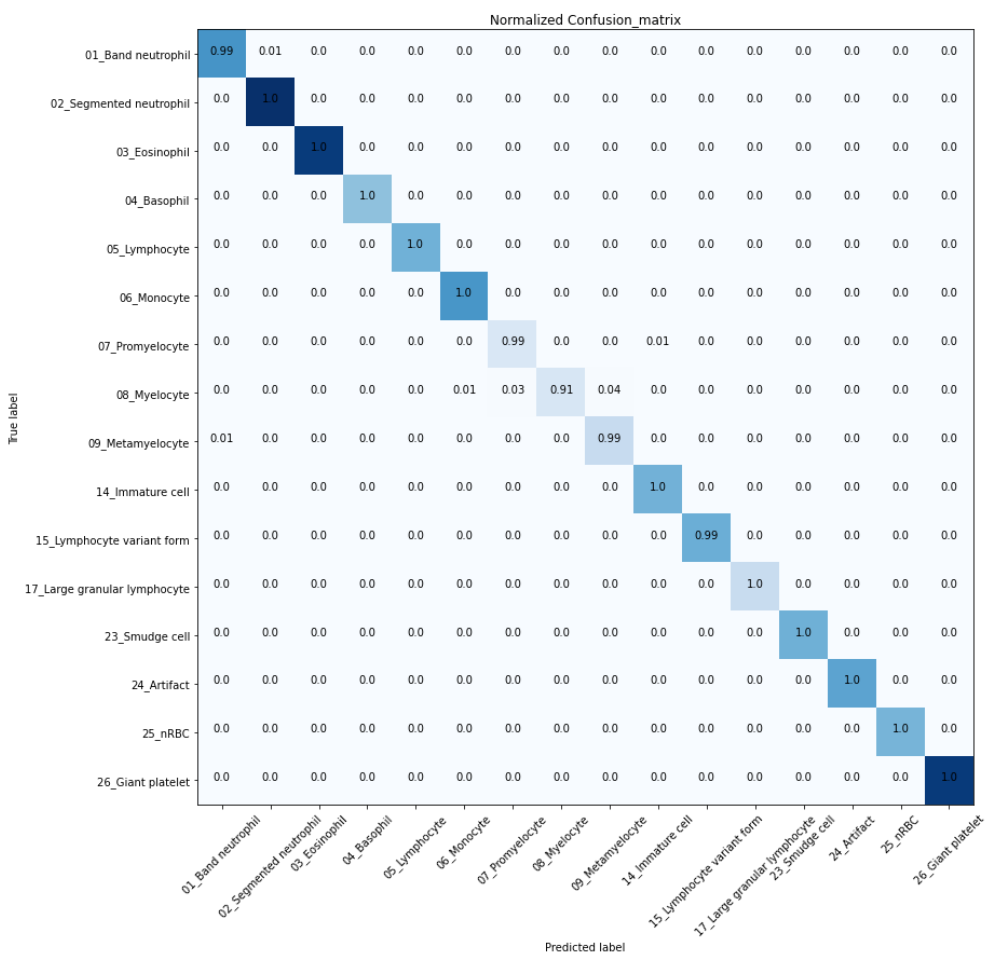


Figure 5.4: Confusion matrix of ViT_L16.

Chapter 6

Discussion

6.1 Discussion and Remaining Challenges

White blood cells (WBCs) are instrumental in diagnosing and monitoring numerous illnesses, emphasizing the critical nature of their accurate detection and classification. Traditional methods of classification, performed manually by trained professionals, bring specific challenges, such as the risk of human error, potential bias, and the time-consuming nature of the task. Consequently, revising and innovating these methods is imperative, seeking improved accuracy, efficiency, and objectivity. Deep learning techniques have emerged as a promising alternative for detecting and classifying WBCs in microscopic images. Over recent years, these techniques have witnessed significant advancements, strengthening their potential role in the medical field. However, the application of such sophisticated techniques has its challenges. The inherent complexity of accurately identifying different types of WBCs is one of the substantial challenges. Differentiating these cell types can be daunting, with the human body's Red Blood Cell (RBC) to white blood cell (WBC) ratio standing at 600:1. Additionally, the presence of diverse WBC types, such as monocytes, eosinophils, basophils, lymphocytes, and neutrophils, each essential in combating infections, introduces further intricacy into the process. Another challenge is the scarcity of annotated datasets of realistic WBC images, particularly those of rare classes. This shortage hinders the advancement of deep learning techniques, stalling progress in refining detection and classification methods. Imbalanced class distributions within WBC datasets further complicate the situation. Some classes contain significantly fewer samples than others, leading to possible detection and classification model bias towards the more abundant types. This imbalance can

skew model accuracy, diminishing reliability for detecting and classifying rarer WBC classes. Recognizing these challenges, our study offers a groundbreaking approach by integrating object detection and vision transformer models. Our solution, a 2-way, 2-stage model leveraging the You Only Look Once (YOLO) and Vision Transformer (ViT) methods, robustly addresses these challenges. This innovative model integrates convolutional and transformer-based approaches for medical imaging tasks, effectively overcoming the limitations of traditional and deep learning methods and demonstrating immense potential. The proposed model’s impressive accuracy rate of 0.99640 in classifying 16 types of WBCs attests to its effectiveness. This hybrid model not only addresses the identified challenges but also exhibits significant potential for clinical applications, including early disease detection and monitoring treatment efficacy.

Despite this study’s considerable advancements and progress, some challenges persist. Although our proposed model boasts a classification accuracy of 0.99640 for 16 types of WBCs, enhancing its performance on rare classes and increasing its robustness against noise and other potential data distortions remains necessary. Securing more diverse and extensive annotated datasets also remains a challenge. Despite training our model on a comprehensive multi-class dataset, achieving greater accuracy might necessitate larger datasets, encompassing a more comprehensive range of WBC classes, especially the rare ones. Lastly, the computational complexity and cost of developing and deploying such deep learning models need consideration. Consequently, the ongoing challenge lies in developing a model that balances performance and computational efficiency. In conclusion, while we have made significant strides in advancing WBC detection and classification using deep learning techniques, substantial work remains to be done. The journey towards enhancing the medical field’s ability to classify WBCs efficiently and accurately is ongoing. The ultimate aim remains to aid in the early diagnosis and treatment of various diseases, underscoring the importance of continued progress in this critical area.

Chapter 7

Conclusion

7.1 Conclusion and Future work

This research proposes an innovative methodology for the identification and categorization. Specifically, it introduces a novel approach for detecting and classifying WBCs utilizing YOLO and ViT techniques. The methodology involves object detection and cropping to identify WBC regions, yielding two separate datasets. We subsequently apply feature extraction and concatenation to the merged WBC and nucleus region datasets before classifying them using our pre-trained ViT model. Remarkably, the model attains an accuracy of 99.640% on 16 classes , including rare classes.

According to our ablation analysis, we obtained the following observations. First, implementing 2-way object detection, which involves detecting WBCs and their respective nucleus, improved accuracy in WBC classification. Second, our analysis revealed that ViT outperforms CNN-based models in WBC classification tasks. The self-attention mechanisms of ViT model the changes in local images and their long-range dependencies. This finally captures the global contextual information within WBC images. Third, our 2-stage model approach leverages on the strengths of object detection and classification, leading to improved overall performance. The object detection stage ensures precise localization of the WBCs, while the subsequent classification stage focuses on accurately classifying the detected cells. These findings suggest that incorporating 2-way object detection and adopting ViT in WBC classification tasks can boost the performance. By implementing these techniques, researchers and practitioners can enhance their understanding of WBCs, which would contribute to disease diagnosis and advance our knowledge of the immune system.

Summary

WBC YOLO-ViT:

2 Way - 2 Stage White Blood Cell Detection and Classification with a Combination of YOLOv5 and Vision Transformer

This groundbreaking study proposes an innovative deep learning-based approach to overcome the longstanding challenges in accurately detecting and classifying white blood cells (WBCs), which play a pivotal role in diagnosing and monitoring various diseases. The researchers identify the limitations of conventional methods and seek to resolve them through modern deep learning techniques. The role of WBCs in the body is intricate, with different types of WBCs, such as monocytes, eosinophils, basophils, lymphocytes, and neutrophils, combating various infections. However, accurately identifying these WBC types within a blood sample is no small feat due to the high ratio of red blood cells (RBCs) to WBCs and the inherent complexity of classifying WBCs.

In response to these challenges, the research team turned to advanced deep learning techniques, leveraging methods like K-means clustering, edge detection algorithms, and convolutional neural networks (CNNs) for feature extraction. Despite these advancements, they observed that deep learning models still encounter significant challenges due to limited annotated datasets and imbalanced class distributions within WBC datasets. To address these obstacles, the researchers proposed a unique 2-way 2-stage model that synergizes the You Only Look Once (YOLO) object detection and Vision Transformer (ViT) classification techniques. This model operates on two types of images: WBC and nucleus images. The study stands out due to its comprehensive dataset encompassing a

broad range of 16 distinct WBC classes, including rarer ones. Upon evaluating the model’s performance, the researchers discovered that ViT outperforms other pre-trained models, achieving an impressive accuracy rate of 99.640% for the 16 WBC classes, even when rare classes were included. However, the model faced difficulties identifying specific WBC types, such as Myelocyte and Metamyelocyte.

Further investigation through ablation analysis solidified the advantages of the proposed model, emphasizing the significance of object detection and the robustness of the ViT architecture. Even without object detection and excluding nucleus images, the ViT model maintained relatively high performance. This underscores the resilience of the proposed model.

In conclusion, this study presents a novel methodology for WBC detection and classification that harnesses the strengths of object detection and classification. By implementing these techniques, the research showcases a potential pathway to enhance disease diagnosis and our understanding of the immune system, ultimately leading to more efficient healthcare practices.

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Appendix A

Abbreviations

WBC YOLO-ViT White Blood Cell with You Only Look Once and Vision Transformer

YOLOV5 You Only Look Once Version 5

WBCs White Blood Cells

RBCs Red Blood Cells

CNNs Convolutional Neural Networks

mAP Mean Average Precision

CNN Convolutional Neural Network

RCNN Region-based Convolutional Neural Network

BCCD Blood Cell Count Dataset

IoU Intersection of Union

ROI Regions of Interest

Appendix B

Source codes for White Blood Cell Detection and Classification with a Combination of YOLOv5 and Vision Transformer

```
## Library
# Libraries and Configurations
import pandas as pd
import numpy as np
import tensorflow as tf
import tensorflow_addons as tfa
import glob, warnings
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, classification_report
import seaborn as sns

warnings.filterwarnings('ignore')
print('TensorFlow Version ' + tf.__version__)

# Define the class_size and Image size
num_classes = 16
IMAGE_SIZE = 224, 224

# 1. ViT B16 Model
vit_model = vit.vit_l16(
    image_size = IMAGE_SIZE,
    activation = 'softmax',
    pretrained = True,
    include_top = False,
    pretrained_top = False,
```

```

        classes = 16)

# 2. Fine-tuning the Model
model = tf.keras.Sequential([
    vit_model,
    tf.keras.layers.Flatten(),
    tf.keras.layers.BatchNormalization(),
    tf.keras.layers.Dense(11, activation = tf.keras.activations.gelu),
    tf.keras.layers.BatchNormalization(),
    tf.keras.layers.Dense(16, 'softmax')
],
    name = 'vision_transformer')

model.summary()

optimizer=tf.keras.optimizers.Adam(learning_rate=0.0001)
model.compile(loss=keras.losses.categorical_crossentropy, optimizer=optimizer,
    ↪ metrics=['accuracy'])

# Save model
filepath = 'models/vit_l16_0G.h5'

# Function to perfome the Kfold
from sklearn.model_selection import KFold
def func_cv(X_train, y_train, cv_itr, n_cv):
    c_matrix_test = np.zeros((num_classes, num_classes), dtype=int)
    cm_agg = np.zeros((num_classes, num_classes), dtype=int)

    res_acc = np.zeros(n_cv)
    res_pre = np.zeros(n_cv)
    res_rec = np.zeros(n_cv)
    res_f1s = np.zeros(n_cv)

    res_acc_ts = np.zeros(n_cv)

```



```

res_pre_ts = np.zeros(n_cv)
res_rec_ts = np.zeros(n_cv)
res_fis_ts = np.zeros(n_cv)

ix = 0

# Split the training set into testing and training sets
for train_ix, test_ix in cv_itr.split(X_train, y_train):
    x_tr, y_tr = X_train[train_ix], y_train[train_ix]
    x_test, y_test = X_train[test_ix], y_train[test_ix]

    # Now we split the training set further into training and validation
    ↪ sets
    inner_kf = KFold(n_splits=10, shuffle=True)
    for inner_train_index, inner_val_index in inner_kf.split(x_tr):
        X_inner_train, X_inner_val = x_tr[inner_train_index],
        ↪ x_tr[inner_val_index]
        y_inner_train, y_inner_val = y_tr[inner_train_index],
        ↪ y_tr[inner_val_index]

    print('-'* 80)
    print('{0}-Training and Testing Data for this fold'.format(ix))
    print(x_tr.shape, y_tr.shape)
    print(x_test.shape, y_test.shape)
    print('-'* 80)
    print('{0}-Training and Validation Data for this fold'.format(ix))
    print(X_inner_train.shape, y_inner_train.shape)
    print(X_inner_val.shape, y_inner_val.shape)
    print('-'* 80)
    print('\n')

    history = model.fit(X_inner_train, y_inner_train,
                        validation_data=(X_inner_val, y_inner_val),

```

```

epochs= 20,
callbacks=[
    tf.keras.callbacks.ReduceLROnPlateau(
        monitor = 'val_accuracy',
        factor = 0.2,
        patience = 10,
        verbose = 1,
        min_delta = 1e-4,
        min_lr = 1e-6,
        mode = 'max'),

    tf.keras.callbacks.EarlyStopping(
        monitor = 'val_accuracy',
        min_delta = 1e-4,
        patience = 15,
        mode = 'max',
        restore_best_weights = True,
        verbose = 1),

    tf.keras.callbacks.ModelCheckpoint(
        filepath = filepath,
        monitor = 'val_accuracy',
        verbose = 1,
        save_best_only = True,
        save_weights_only = False,
        mode = 'max')
])

```

```

# serialize the model to disk
print("[INFO] Saving Model...")
# load a saved model
saved_model = tf.keras.models.load_model(
    filepath,
    custom_objects=None,
    compile=True,

```

```

        options=None
    )

    # START ---TRAINING AND VALIDATION PART---
    y_pred_ = saved_model.predict(X_inner_val)
    y_pred1 = []
    y_pred2 = []

    for i in y_pred_:
        y_pred1.append(np.argmax(i))
    for j in y_inner_val:
        y_pred2.append(np.argmax(j))

    y_pred = np.array(y_pred1)
    y_pred_ts = np.array(y_pred2)

    print('\n')
    print('-'* 40)
    print(time.strftime('%X %x %Z'))
    print('{0}-CV Confusion Matrix'.format(ix))
    print('-'* 40)

    cr = classification_report(y_pred_ts, y_pred, digits=3,
    ↪ output_dict=True)

    #Training
    res_acc[ix] = cr['accuracy']
    res_pre[ix] = cr['macro avg']['precision']
    res_rec[ix] = cr['macro avg']['recall']
    res_f1s[ix] = cr['macro avg']['f1-score']

```

```

print('\n')
print('-'* 80)
print('Training ')
print('Accuracy:\t\t {0:.5f}'.format(cr['accuracy']))
print('Avg Precision:\t\t {0:.5f}'.format(cr['macro avg']['precision']))
print('Avg Recall:\t\t {0:.5f}'.format(cr['macro avg']['recall']))
print('Avg F1-score:\t\t {0:.5f}'.format(cr['macro avg']['f1-score']))
# END ---TRAINING AND VALIDATION PART---

```

```

# START ---TESTING PART---

y_pred_test = saved_model.predict(x_test)
y_pred_test1 = []
y_pred_test2 = []

for i in y_pred_test: # X_test
    y_pred_test1.append(np.argmax(i))
for j in y_test: # y_test
    y_pred_test2.append(np.argmax(j))

y_pred_t = np.array(y_pred_test1)
y_pred_tt = np.array(y_pred_test2)

```

```

# Testing Classification
c_matrix_test = confusion_matrix(y_pred_tt, y_pred_t)

cm_agg = np.add(cm_agg, c_matrix_test)
cr_test = classification_report(y_pred_tt, y_pred_t, digits=3,
↪ output_dict='true')

```

```

#Testing

```

```

res_acc_ts[ix] = cr_test['accuracy']
res_pre_ts[ix] = cr_test['macro avg']['precision']
res_rec_ts[ix] = cr_test['macro avg']['recall']
res_f1s_ts[ix] = cr_test['macro avg']['f1-score']


print('\n')
print('-'* 80)
print('Testing')
print('Accuracy_test:\t\t\t {0:.5f}'.format(cr_test['accuracy']))
print('Avg Precision_test:\t\t {0:.5f}'.format(cr_test['macro
↪ avg']['precision']))
print('Avg Recall_test:\t\t {0:.5f}'.format(cr_test['macro
↪ avg']['recall']))
print('Avg F1-score_test:\t\t {0:.5f}'.format(cr_test['macro
↪ avg']['f1-score']))
print('')
print('-'* 80)
print('\n\n')
# END ---TESTING PART---

ix = ix + 1


print("*" * 20 + "END" + "*" * 20)
print('Aggregated Confusion Matrix for Training ')
print('-'* 80)
print('Aggregated avg Accuracy:\t {0:.5f}'.format(np.mean(res_acc)))
print('Aggregated avg Precision:\t {0:.5f}'.format(np.mean(res_pre)))
print('Aggregated avg Recall:\t\t {0:.5f}'.format(np.mean(res_rec)))
print('Aggregated avg F1-score:\t {0:.5f}'.format(np.mean(res_f1s)))
print('')

print('-'* 80)

```

```

print('Aggregated Confusion Matrix for Testing')
print('-'* 80)
print('Aggregated AVG Accuracy_test:\t {0:.5f}'.format(np.mean(res_acc_ts)))
print('Aggregated AVG Precision_test:\t
↪ {0:.5f}'.format(np.mean(res_pre_ts)))
print('Aggregated AVG Recall_test:\t {0:.5f}'.format(np.mean(res_rec_ts)))
print('Aggregated AVG F1-Score_test:\t {0:.5f}'.format(np.mean(res_f1s_ts)))
print('')
print('-'* 80)
print('')

def plot_confusion_matrix(cm_agg, classes,
                           normalize=True,
                           title='Normalized confusion matrix',
                           cmap=plt.cm.Blues):

    plt.imshow(cm_agg, interpolation='nearest', cmap=cmap)
    plt.title(title)
    plt.colorbar()
    tick_marks = np.arange(len(classes))
    plt.xticks(tick_marks, classes, rotation=45)
    plt.yticks(tick_marks, classes)

    if normalize:
        cm_agg = np.around(cm_agg.astype('float') / cm_agg.sum(axis=1)[:],
                            ↪ np.newaxis], decimals= 2)
        print()
        # print("Normalized confusion matrix")
    else:
        print()
        print('Confusion matrix, without normalization')

    # print(cm_agg)

    thresh = cm_agg.max() / 2.

```

```

for i, j in itertools.product(range(cm_agg.shape[0]),
    ↪ range(cm_agg.shape[1])):
    plt.text(j, i, cm_agg[i, j],
             horizontalalignment="center",
             color="black" if cm_agg[i, j] > thresh else "black")

plt.tight_layout()
plt.ylabel('True label')
plt.xlabel('Predicted label')

cm_plot_labels = ['01_Band neutrophil', '02_Segmented neutrophil',
    ↪ '03_Eosinophil', '04_Basophil', '05_Lymphocyte', '06_Monocyte',
             '07_Promyelocyte', '08_Myelocyte', '09_Metamyelocyte',
    ↪ '14_Immature cell', '15_Lymphocyte variant form',
             '17_Large granular lymphocyte', '23_Smudge cell',
    ↪ '24_Artifact', '25_nRBC', '26_Giant platelet']

fig, ax = plt.subplots(figsize=(15, 15))
#plot_confusion_matrix(your_model, X_test, y_test, ax=ax)
plot_confusion_matrix(cm_agg=cm_agg, classes=cm_plot_labels,
    ↪ title='Normalized Confusion_matrix')

# Library
from sklearn.model_selection import StratifiedKFold, KFold
# Define number of cross validation
n_cross_valid = 10
kf = KFold(n_splits=n_cross_valid, shuffle=True, random_state=44)

import os
import sys
import time
#org_stdout = sys.stdout
##f = open('Regression_Exp_001' + '.txt', 'w')
#sys.stdout = f

```

```

print('\n')
print("*" * 20 + "START" + "*" * 20)
print('*'*50)
print("VISION TRANSFORMERS (ViT_B16)")
print(time.strftime('%X %x %Z'))

print('*'*50)
print('\n')
func_cv(X_train,y_train_wbc, kf, n_cross_valid)
func_cv(x_train,y, skf, n_cross_valid)

f.close()
sys.stdout.close()
sys.stdout = org_stdout

```


Acknowledgements

First and foremost, I humbly express my gratitude to the Almighty God, whose boundless blessings have been instrumental in the success of my research endeavors. His benevolence has guided me throughout my academic pursuits, and I am immensely grateful.

Second, I would like to extend my sincere appreciation to Prof. Jiyoung Woo from the Department of AI and Big Data at Soonchunhyang University, Asan 31538, South Korea. Her steadfast support, invaluable advice, and guidance have been instrumental in my research journey. Prof. Woo exceptional wisdom, vast knowledge, and a firm commitment to excellence have inspired and motivated me immensely. I am deeply fortunate to have him as my Master's degree advisor.

I am profoundly thankful to my parents, who have been a constant source of motivation for me to contribute positively to society. I owe a debt of gratitude to my siblings and other relatives, as I couldn't devote as much time to them during my Master's studies. This work would not have come to fruition without their selfless support and generosity.

Lastly, I would like to express my appreciation to my fellow students in the Department of Future Convergence Technology, whose support and guidance have propelled me forward. I am grateful to all my friends and well-wishers who have provided moral and financial support throughout this journey, directly and indirectly. Their encouragement has been invaluable, and I am truly grateful for their presence in my life.

